

NITHISH KRISHNABHARATHI GNANI

Scania Thesis Worker | Experienced Researcher in Autonomous Systems and Applied AI | Stockholm, Sweden

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WORK EXPERIENCE

Master's Thesis Worker 2026 - Present
Scania *Södertälje, Sweden*
Building data-driven models from real truck telemetry to quantify driver-related impact on energy consumption and support the development of future eco-driving assistance systems, with grouped evaluation across diverse operating contexts.

Technical Associate [R&D] 2019 - 2024
Indian Institute of Science (IISc) | Centre for Networked Intelligence (CNI)  *Bangalore, India*
Worked on open-ended research problems across robotics, sensing, networking, and cyber-physical systems, resulting in journal and IEEE publications, including a Best Paper Award.

- Developed ML and signal-processing workflows for localization, anomaly assessment, adaptive decision-making, and operator-intent modelling using real and simulated sensor data.
- Built experimental pipelines spanning sensing hardware, telemetry capture, embedded systems, and downstream models for deployment-minded research.
- Contributed to projects with academic and industry collaborators across aerospace, robotics, and communication systems.

Team Leader, Maintenance [Manufacturing Plant] 2017 - 2018
Coca-Cola (HCCB Pvt Ltd) *Hyderabad, India*
Managed a team of 20+ associates to improve operational reliability of plant utility systems and coordinate data-driven maintenance routines.

EDUCATION

Master of Science, Intelligent Software for Autonomous Systems Sep 2024 - Jun 2026
KTH Royal Institute of Technology, Sweden & Université Côte d'Azur, France
EIT Digital dual master with Minor in Innovation and Entrepreneurship
Relevant coursework: Artificial Intelligence, Applied Estimation, Computer Vision, Sensor Fusion, Dependable Autonomous Systems

Bachelor of Technology, Mechanical Engineering 2013 - 2017
National Institute of Technology Karnataka (NITK), India

SELECTED PUBLICATIONS | MORE

Utilizing Operator Intent for Haptic Teleoperation under High Latencies 
IEEE Transactions on Mobile Computing, 2025

EdgeP4: A Shared Edge Computing Platform for Multi-Tenant Haptic Teleoperation 
IEEE INFOCOM CNERT, 2024

Best Paper Award

Towards a TSN-DetNet Intercity Testbed for Tactile Cyber-Physical Systems 
IEEE INFOCOM CNERT, 2024

Sensor Identification via Acoustic Physically Unclonable Function 
Digital Threats: Research and Practice, 2023

Judicious Data Management for Sustaining an Energy Harvesting Sensor Node 
Concurrency and Computation: Practice and Experience, 2021

RoboTwin: A Robotic Teleoperation Framework Using Digital Twins 
Preprint, 2024

SELECTED PROJECTS | MORE

Driving Data Analysis and Driver Support System (Scania)

- Built context-only and driver-aware energy-consumption models on real truck telemetry to quantify driver impact under similar operating conditions.

- Designed grouped train/validation/eval workflows across context buckets and used SHAP analysis to show that driver-control signals add explanatory value beyond context alone.
- The ongoing thesis contributes toward future real-time driver-support logic for eco-driving assistance.

Teleoperation Across Cities and Continents (IISc/TU Delft) [\[🔗\]](#)

- Built a 400 km intercity tactile cyber-physical system testbed for bilateral haptic teleoperation and later extended the work to intercontinental experiments with TU Delft.
- Developed EdgeP4, a shared edge-computing platform for teleoperation, reducing feedback-correction latency to below 100 microseconds and network load by up to 99.9% [\[paper\]](#).
- Developed operator-intent mechanisms for high-latency teleoperation, improving responsiveness under adverse network conditions and leading to a publication in *IEEE Transactions on Mobile Computing* [\[paper\]](#).
- Contributed to RoboTwin, a digital-twin framework for teleoperation that reduced data rate by 25x while preserving task performance [\[paper\]](#).

Aircraft IoT Data Management (IISc/Boeing) [\[🔗\]](#)

- Built a real-time monitoring and anomaly-assessment pipeline for aircraft cabin sensor data using MATLAB/Simulink time-series models.
- Developed adaptive storage logic for critical sensor streams, achieving 96.6% storage savings while preserving faithful reconstruction [\[journal paper\]](#).
- Integrated the pipeline with the TICK stack using InfluxDB, Kapacitor, and Chronograf for telemetry processing and storage.

Acoustic Localization and Device Identification (IISc/Boeing) [\[🔗\]](#)

- Built a localization pipeline using fingerprinted time-difference-of-arrival features from embedded acoustic sensor nodes.
- Applied kNN inference to held-out test data and achieved 98% localization accuracy; the work later informed a journal paper on acoustic PUF-based device identification [\[journal paper\]](#).

SKILLS

Programming: Python, MATLAB, SQL, Shell, C, C++

ML and Analytics: Time-series modelling, classification, regression, feature engineering, reinforcement learning

Systems: Linux, Git, Docker, embedded sensing, telemetry pipelines, experimental evaluation

Domains: Autonomous systems, vehicle telemetry, IoT sensor data, robotics, cyber-physical systems

Languages: English (native), French (TCF B1)

INTERESTS AND ACTIVITIES

- Instructor for advanced courses on 5G, programmable networking, and cyber-physical systems for NSE TalentSprint, IISc, and Power Grid Corporation of India Limited
- Designing and 3D printing functional products and custom hardware parts [\[🔗\]](#)
- Volunteer interview coaching and mentoring for underprivileged students through Yuwa School, India [\[🔗\]](#)